INCH-POUND
MIL-DTL-19865D(SH)
8 September 2021
SUPERSEDING
MIL-A-19865C(SH)
20 March 1990

DETAIL SPECIFICATION

AIR CONDITIONER, MECHANICALLY REFRIGERATED, SEAWATER COOLED

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanically refrigerated, seawater cooled air conditioners for Naval shipboard use in air circulation, air cooling, and dehumidification.

1.2 Classification. The air conditioners are of the following types, arrangements, service voltage, rated capacities, and shock grades as specified (see 6.2).

1.2.1 Types.

a. Type I - Drip-proof protected (Part or identifying number [PIN] Code 01)

b. Type II - Explosion-proof enclosed (PIN Code 02)

1.2.2 Air discharge plenum arrangements.

a. Arrangement A - With air discharge plenum (PIN Code A)

b. Arrangement B - Without air discharge plenum (PIN Code B)

1.2.3 Service voltage.

a. Service 1 - 440 volts alternating current (VAC) 60 hertz (Hz) (PIN Code 1)

1.2.4 Rated capacities.

a. Size 2 2 tons of refrigeration [TOR] capacity (PIN code 2)

b. Size 3 3 TOR (PIN Code 3)

c. Size 5 5 TOR (PIN Code 5)

d. Size 7 7.5 TOR (PIN Code 7)

1.2.5 Shock ratings in accordance with MIL-DTL-901.

a. Grade A - (PIN Code A)

b. Grade B - (PIN Code B)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4120

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1.3 Part or identifying number (PIN). The PINs to be used for seawater cooled air conditioners acquired to this specification are created as follows:

M	19865	-	XX	X	X	-	X	X
Prefix for Military Specification	Specification Number	-	Type I (PIN Code 01) Type II (PIN Code 02) (see 1.2.1)	Plenum arrangement (PIN Code A or B) (see 1.2.2)	Service voltage (PIN Code 1) (see 1.2.3)	-	Unit Size (PIN Code 2, 3, 5, or 7) (see 1.2.4)	Shock grade (PIN Code A or B) (see 1.2.5)

Example PIN: M19865-02A1-7B is an explosion-proof enclosed, air plenum equipped, 440 VAC, 7.5 TOR, grade B shock rated, mechanically refrigerated, seawater cooled air conditioner.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

TT-C-490 - Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

TT-P-645 - Paint, Alkyd Type, Formula Number 84

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-1183 - Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Application where a High Degree of Reliability is Required, General Specification for

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-P-15024/5 - Plate, Identification

MIL-DTL-15090 - Enamel, Equipment, Light Gray (Navy Formula No. 111)

MIL-C-15726 - Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire

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MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)

MIL-M-17059 - Motors, 60-Cycle, Alternating-Current, Fractional H.P. (Shipboard Use)

MIL-DTL-17060 - Motors, Alternating Current, Integral-Horsepower, Shipboard Use

MIL-B-17931 - Bearing, Ball, Annular, for Quiet Operation

MIL-DTL-18001 - Anodes, Sacrificial Zinc Alloy

MS18229 - Plug for O-Ring Gasket

MIL-A-19521 - Anode Retaining Support Plugs and Anode Selection and Installation Design Criteria for Shipboard Condensers and Heat Exchangers

MIL-PRF-20042 - Flanges, Pipe, Bronze (Silver Brazing)

MIL-G-21610 - Gaskets, Heat Exchanger, Various Cross Section Rings, Synthetic Rubber

MIL-T-22214 - Tube, Condenser and Heat Exchanger with Integral Fins (UNS Alloy Nos. C71500, C70600, C12200)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment

MIL-STD-882 - System Safety

MIL-STD-889 - Galvanic Compatibility of Electrically Conductive Materials

MIL-STD-1472 - Human Engineering

MIL-STD-1474 - Noise Limits

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume 1

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

SNAP - Significant New Alternatives Policy

(Copies of this document are available online at https://www.epa.gov.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-841499 - Strainers, Y-Type Assembly and Details

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

T9070-AJ-DPC-120/3010 - Shock Design Criteria for Surface Ships

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR CONDITIONING, HEATING, AND REFRIGERATION INSTITUTE (AHRI)

AHRI 210/240-2023 - Performance Rating of Unitary Air-conditioning & Air-source Heat Pump Equipment

(Copies of this document are available online at http://www.ahrinet.org.)

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ABMA 4 - Tolerance Definitions and Gauging Practices for Ball and Roller Bearings

(Copies of this document are available online at www.americanbearings.org/.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 15 - Safety Standard for Refrigeration Systems

ASHRAE 34 - Designation and Safety Classification of Refrigerants

ASHRAE 37 - Methods of Testing for Rating Electrically Driven Unitary Air-Conditioning and Heat Pump Equipment

(Copies of these documents are available online at www.ashrae.org/.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://webstore.ansi.org.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC, Section VIII, Division 1 - Design & Fabrication of Pressure Vessels

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A276/A276M	-	Standard Specification for Stainless Steel Bars and Shapes
ASTM A1011/A1011M	-	Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
ASTM B16/B16M	-	Standard Specification for Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B75/B75M	-	Standard Specification for Seamless Copper Tube
ASTM B148	-	Standard Specification for Aluminum Bronze Sand Castings
ASTM B209	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

(Copies of these documents are available online at www.astm.org.)

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C504 - Standard for Rubber-Seated Butterfly Valves

(Copies of this document are available online at www.awwa.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA HP 3	-	Insulated High-Temperature Hook-Up Wire; Types ET (250 Volts), E (600 Volts), and EE (1000 Volts)
NEMA MG 1	-	Motors and Generators

(Copies of these documents are available online at www.nema.org.)

SAE INTERNATIONAL

SAE J1086	-	Numbering Metals and Alloys
SAE J1926/1	-	Connections for General Use and Fluid Power – Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing – Part 1: Threaded Port with O-Ring Seal in Truncated Housing, Standard

(Copies of these documents are available online at www.sae.org.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 873	-	Standard for Temperature-Indicating and Regulating Equipment
UL 1995	-	Standard for Safety, Heating and Cooling Equipment
UL 2182	-	Standard for Safety, Refrigerants

(Copies of these documents are available online at www.ul.com.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.2.2.

3.2 General. Air conditioners covered by this specification shall consist of a condensing unit section, an air handling/evaporator section, and a motor controller. An air discharge plenum shall be included, when specified (see 1.2 and 6.2). These air conditioners shall be self-contained and shall perform as specified herein when connected to cooling water, drainage, and electric power in accordance with UL 1995 and AHRI 210. The refrigerant systems shall be constructed in accordance with the requirements of ASHRAE 15, except for the condenser, which shall be designed in accordance with ASME Boiler and Pressure Vessel Code Section VIII, Division I but shall not be stamped. Welding shall be in accordance with S9074-AR-GIB-010/278, except radiography is not required.

3.3 Reliability. The minimum acceptable mean time between failures for each air conditioner shall be 20,000 operating hours.

3.4 Materials. Materials shall be as specified in table I and herein for the systems and components the appropriate application of which is accepted by the Technical Authority in conjunction with first article tests. The use of alternate materials shall not be permitted unless such alternate materials have been successfully tested during the air conditioner first article tests or have been accepted as alternate materials by the Technical Authority. The materials shall be free from any defects that might affect the serviceability or appearance of the finished product.

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TABLE I. Seawater cooled air conditioner material list.

Corrosion-resisting steel	ASTM A276/A276M UNS S30400, S31600, or S34700 alloys. These corrosion-resisting steels, when fabricated by any method that tends to reduce corrosion-resisting properties, shall be normalized.
Condenser inlet and outlet piping	Schedule 40 copper-nickel
Zinc anodes	MIL-DTL-18001
Condenser gaskets	MIL-G-21610, type I
Condenser tube sheets	MIL-C-15726 90-10 copper-nickel
Condenser heads	ASTM B61/B61M valve bronze; or ASTM B148 nickel-aluminum-bronze, C95800 alloy, heat treated per AWWA C504
Condenser tubes	MIL-T-22214 90-10 copper-nickel
Condenser vent and drain fittings and anode holders	QQ-N-281 70-30 nickel-copper
Condenser bolting material	MIL-DTL-1222 grades 400, 405, 510, 544, 632, 655, or 661
Cooling coil (evaporator) tubes	ASTM B75/B75M alloy C12200
Water-regulating valve body	ASTM B61 valve bronze alloy C92200, or ASTM B148 alloy C95800
Water-regulating valve internal metal parts subject to corrosion or erosion	QQ-N-281 70-30 nickel-copper
Water-regulating valve piping	MIL-T-16420 alloy C70600, type I, class 200, grade 1
Water pipe fittings and seawater service connections	ASTM B61 90-10 copper-nickel or valve bronze; or QQ-N-280 70-30 copper-nickel
Refrigerant pipe fittings	ASTM B16/B16M
Refrigerant piping and evaporator condensate drain tube	ASTM B75/B75M alloy C12200
Cabinet enclosure	ASTM B209 alloy A95052, temper H-32 ¼ hard; or ASTM A1011/A1011M low carbon steel or equivalent

NOTES:

1. SAE J1086 Unified Numbering System (UNS) designation. Reference to a specific UNS designation permits the use of any one of the many alloy specifications and standards listed in SAE J1086 for that UNS number.
2. Stainless 304 alloys shall not be used for wetted surfaces.

3.4.1 Prohibited materials. Air conditioners shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.4.2 Corrosion protection. Corrosion-resisting steel, copper, brass, bronze, chromium, copper-nickel, and copper-nickel alloys as specified herein are considered corrosion-resisting materials. Where corrosion-resisting steel is used, it shall be type UNS S30400, S31600, or S34700 in accordance with ASTM A276/A276M. S30400 shall not be used in seawater exposed surfaces. Corrosion-resisting steel, when fabricated by any method that tends to reduce corrosion-resisting properties, shall be normalized to restore those properties before being used in the assembly of any unit (see 4.4). Parts other than corrosion-resisting materials shall be protected against corrosion by the use of chemicals, electrolytic processes, plating, or paints and enamels.

3.4.3 Fastenings or fittings. Bolts, nuts, studs, pins, screws, and such other fastenings or fittings shall be of corrosion-resisting material. Self-tapping screws with machine screw heads may be used in the cabinet assembly. Sheet metal screws with sheet metal threads shall not be permitted.

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3.4.4 Galvanic corrosion. Direct contact of electrolytically dissimilar materials shall comply with MIL-STD-889 to prevent galvanic corrosion.

3.4.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically permitted.

3.5 Construction. Unless otherwise specified (see 6.2), each air conditioner shall consist of a condensing unit, an air handling/evaporator section, and an air discharge plenum (see 6.5.1). Every component, including interconnecting wiring and piping, shall be enclosed within the cabinet, except where external mounting is permitted by this specification. The air conditioner shall be factory assembled, complete, self-contained, with the refrigerant system and components dehydrated and charged with the operating quantities of refrigerant and oil. The air conditioner shall be ready for operation after removal of the shipping protection, opening of valves, adjustments of belts, and connection to services. Valves, controls, and equipment subject to service and repair shall be readily accessible for servicing through removable parts. Where panels are provided for access to equipment, machine screws may be used for fastening.

3.5.1 Refrigerant system. Every component, including piping, controls, and accessories of the refrigerant system, shall be constructed for and compatible with the type and quantity of refrigerant used. Each unit shall be delivered uncharged with refrigerant. Equipment shall use refrigerants that are hydrofluorocarbon (HFC) as a minimum, that are non-ozone depleting, and that are classified as Safety Group A1 in ASHRAE 34. Safety Group A1 includes only those refrigerants with no flame propagation and low toxicity. See ASHRAE 34, table 1, for Safety (Toxicity and Flammability) Group Classifications. In addition, refrigerants shall comply with UL 2182 and shall be listed as acceptable under the Environmental Protection Agency's (EPA) Significant New Alternatives Policy (SNAP) Program.

3.5.2 Safety. System safety requirements shall be in accordance with MIL-STD-882. To ensure maximum protection for the operating and maintenance personnel against hazards and accidents, the air conditioner shall incorporate the following safeguards and protective features:

a. Exposed moving and rotating parts shall be covered by removable guards in accordance with MIL-STD-1472.

b. Components with exposed operating surface temperatures in excess of 52 °C (125 °F) that are within easy grasp, or that are susceptible to being inadvertently touched in everyday operating tasks, shall have those high temperature surfaces insulated or covered with metallic guards.

c. Exposed edges shall be generously rounded and made smooth to prevent cutting edges and sharp corners.

d. Insulated connectors and plugs of the electric current carrying conductors shall be constructed in such a way that they will not constitute an electrical shock or residual discharge hazard when they are disconnected for maintenance and repair.

e. Controls and switches shall be recessed from the outer sides of the enclosure to preclude accidental contact or breakage.

3.5.3 Ease of maintenance. The air conditioner shall have a mean time to repair of 2 hours and a mean time between corrective maintenance of 20,000 hours. Air conditioners shall be constructed so they may be easily maintained, disassembled, repaired, and reassembled (see 4.6.8). Maintenance operations shall be straightforward and simple with proper regard given to the fact that maintenance often will be performed under adverse conditions with the ship at sea, far from normal sources of supply, by maintenance personnel who are not seasoned mechanics. Criteria for ease of maintenance shall include parts standardization and interchangeability as specified in 3.5.3.1 and 3.5.3.2.

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3.5.3.1 Parts standardization. The air conditioner construction shall standardize, as far as practicable, those components, fasteners, and items used in manufacture so as to reduce the number of sizes and kinds of items that are generally similar. To the maximum extent possible, standard, commercially available parts shall be used. For example, an air conditioner incorporating grade 8 ¼-20 UNC-2-inch bolts in one location and grade 5 ¼-20 UNC 2-inch bolts in another location should standardize its design to use grade 8 ¼-20 UNC-2-inch bolts. O-ring material shall be standardized to the maximum extent possible. O-ring diameters and sizes shall be those commonly available from industrial supply houses. Parts shall not be standardized where standardization will result in any improper use.

3.5.3.2 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.6 Rated capacity. Air conditioners shall be supplied in standard capacities as specified (see 6.2). The standard capacities are the following:

Size 2 - minimum capacity 2 TOR (24,000 British thermal units [Btus] per hour)

Size 3 - minimum capacity 3 TOR (36,000 Btus per hour)

Size 5 - minimum capacity 5 TOR (60,000 Btus per hour)

Size 7 - minimum capacity 7.5 TOR (90,000 Btus per hour)

3.7 External piping system. The construction of the piping system for shock shall include inlet and outlet condenser water piping of the proper diameter, schedule 40 copper-nickel pipe, extending horizontally 12 inches (305 millimeters) outside of the air conditioner cabinet, cantilevered from the air conditioner inlet and outlet water connections. Construction shall also include the proper size and type of drain piping attached to, and extended from, the drain connection of the air conditioner. The drain piping shall extend through the cabinet down to the deck level. It may be attached to the air conditioner cabinet or deck.

3.8 Dimensions. Maximum dimensions of the air conditioner cabinets including mounting brackets and plenums shall be as shown in table II.

TABLE II. Cabinet and plenum dimensions.

Size	Height (in)(mm)	Width (in)(mm)	Depth (in)(mm)
Cabinets without plenum			
2	65 (89)	36 (91)	22 (56)
3	65 (89)	40 (102)	24 (61)
5	65 (89)	45 (114)	25 (64)
7	65 (89)	48 (122)	25 (64)
Plenums			
2	10 (25)	36 (91)	22 (56)
3	10 (25)	40 (102)	24 (61)
5	10 (25)	45 (114)	25 (64)
7	10 (25)	48 (122)	25 (64)

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3.9 Air conditioner components.

3.9.1 Compressor. The compressor shall be of the open type for service 2 and 3 air conditioners and of the hermetic type for service 1 air conditioners. Provision shall be made for lubrication of the rubbing and wearing surfaces including operation under ship motion specified in 3.16. Compressors shall be provided with a crankcase heater in accordance with UL 1995. Crankcase heaters shall be replaceable without having to remove oil or refrigerant from the compressor and shall be arranged to provide compressor oil heating before start-up and at any time the compressor is in the off cycle. Compressors shall be provided with suction and discharge compressor service shutoff valves and means for charging.

3.9.1.1 Open compressor. The open compressor shall be of the single acting, reciprocating type. The shaft seal and main bearings shall be replaceable in their entirety without the necessity of replacing or refurbishing the crankshaft or crankcase. Compressor speeds for open compressors shall not exceed 1800 revolutions per minute (r/min).

3.9.1.2 Hermetic compressor. The hermetic compressor shall be of the reciprocating or scroll type. Provision shall be made for lubrication of all rubbing and wearing surfaces including operation under ship motion (see 3.16).

3.9.2 Condenser. The condenser shall be seawater cooled and constructed in accordance with the criteria shown in table III and as follows:

a. The condenser shall be sized so that the compressor motor shall not be overloaded when the air conditioner is experiencing overload conditions (see 4.6.6.1).

b. The condenser shall be a shell and tube construction with water (seawater) in the tubes and refrigerant in the shell and shall have an even number of water passes. The condenser shall be mechanically cleanable in place with removable water boxes or heads to permit tubes to be examined, cleaned, or replaced as necessary. The condenser shall be cleanable from either side of the cabinet. Means shall be provided to remove entrained air from each inlet pass. Condenser heat drains shall be provided.

c. Zinc anodes in accordance with MIL-A-19521 shall be installed in the condenser heads (seawater inlet side). The zinc material shall be in accordance with MIL-DTL-18001. The anodes shall be replaceable without removing the condenser heads.

d. Condenser head depth, measured normal to the tube sheet, shall be not less than one half of the inside diameter (id) of the head measured parallel to the tube sheet. Head volume shall be maximized by having a cylindrical rather than a hemispheric shape. The internal shape of the water box shall be streamlined to minimize turbulence. In multi-pass condensers, the joint between the partition edge and the face of the tube sheet shall be provided with a gasket in accordance with type 1 of MIL-G-21610. The condenser heads shall be constructed of valve bronze in accordance with ASTM B61, or nickel-aluminum-bronze in accordance with ASTM B148, alloy C95800, heat treated at 1200 °F (649 °C) for 1 hour to inhibit de-aluminization as prescribed in AWWA C504. The tubes shall be ¾ inch (19 millimeters) outside diameter (od), constructed of copper-nickel (90-10) in accordance with MIL-T-22214.

e. The condenser shall be provided with a means for purging air and non-condensable gases and a relief valve to prevent over-pressurization of the refrigerant side.

f. The tube sheets shall be constructed of copper-nickel (90-10) in accordance with MIL-C-15726 and shall be a minimum of 1 inch (25 millimeters) thick. The tube sheets shall be machined and grooved to accommodate tubes expanded at both ends of the condenser. Tube holes shall be deburred and rounded off to a radius of 1/16 inch (1.6 millimeters) to remove the corner, except that on the water inlet end of the tubes on the outer (water) side of the tube sheet, the tube holes shall be belled and tubes finished flush with the tube sheet.

g. The tubes shall be extruded fin type with a minimum wall thickness, at the root of the fin portion, of 0.049 inch (1.2 millimeters). This id of the tubes shall not be corrugated, finned, or otherwise modified to enhance heat transfer.

h. Drain and vent fittings and zinc anode holders shall be nickel-copper (70-30) in accordance with QQ-N-281. Drain and vent fittings and anode holders shall be O-ring plug straight thread in accordance with MS 18229 to suit a boss tapped in accordance with SAE-J1926/1.

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i. Bolting material, including bolts, studs, and nuts, used to fasten water heads shall be in accordance with grades 400, 405, 510, 544, 632, 655, or 661 of MIL-DTL-1222.

TABLE III. Air conditioner rating conditions.

Operating parameter	Condition
Maximum seawater design pressure	200 lb/in^2 (1379 kilopascals [kPa])
Seawater entering pressure	35 to 50 lb/in^2 (241 to 345 kPa)
Seawater entering temperature	95 °F (35 °C)
Seawater entering temperature, overload condition	100 °F (38 °C)
Seawater velocity through condenser tubes	6.0 feet per second (1.8 meters per second) (maximum)
Refrigerant condensing temperature	115 °F (46 °C) (maximum)
Ambient temperature (entering air)	80 °F (27 °C) dry bulb 67 °F (19 °C) wet bulb
Ambient temperature (entering air, overload condition)	100 °F (38 °C) dry bulb 85 °F (29 °C) wet bulb
Cooling effect ratio (sensible cooling effect to net cooling effect)	65 to 70 percent

Air quantity and static pressure		
Air conditioner size	**Air quantity (minimum)**	**External static pressure**
Size 2 air conditioner	600 feet^3/minute (17 meters^3/minute)	1.0-inch water (0.25 kPa)
Size 3 air conditioner	900 feet^3/minute (25 meters^3/minute)	1.0-inch water (0.25 kPa)
Size 5 air conditioner	1500 feet^3/minute (42 meters^3/minute)	1.0-inch water (0.25 kPa)
Size 7 air conditioner	2250 feet^3/minute (64 meters^3/minute)	1.0-inch water (0.25 kPa)

3.9.3 Liquid receiver. The liquid receiver shall have an internal volume at least 25 percent greater than the volume of the complete refrigerant charge. The receiver shall contain an outlet shutoff valve and pressure relief device in accordance with UL 1995. Where a fusible plug is furnished, the plug shall melt at a temperature not less than 250 °F (127 °C) and not greater than 300 °F (149 °C). The receiver may be made integral with the condenser if the above requirements are met. If the receiver is integral with the condenser, a fusible plug cannot be used.

3.9.4 Cooling coil. The cooling coil (evaporator) shall be of finned-tube construction and shall be composed of copper tubes in accordance with ASTM B75/B75M, alloy C12200, and shall have nonferrous metal fins. Fins shall be firmly bonded to the tube. The fins shall be at least 0.010 inch (0.25 millimeter) thick, with fin spacing 10 to 14 fins per inch. Fins shall have drawn collars at all tube holes. Tube sheets and tube support plates shall include copper ferrules for tube penetrations. Intermediate tube support plates shall be used and the tube length between tube supports shall be not greater than 24 inches (610 millimeters). Fittings and joints shall be silver brazed. A drip pan and drain for collecting the condensate shall be furnished (see 3.9.9).

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3.9.5 Water-regulating valves. A water-regulating valve shall be provided at the outlet to each condenser. The valve shall be direct-acting or pilot-controlled and actuated by condenser gas pressure to modulate the flow of water required for the condenser. The valve shall be selected to regulate the water flow from shutoff to required capacity within a maximum operating gas pressure rise of 40 lb/in^2 (276 kPa). The valve shall be adjustable within an operating range of 180 to 260 lb/in^2 (1241 to 1793 kPa) refrigerant gas pressure. The capacity of the water-regulating valve shall at least equal the water requirement of the condenser, based on available inlet water pressure of 35 lb/in^2 (241 kPa) and a pressure drop across the valve of not more than 10 lb/in^2 (69 kPa). The valve size shall be not less than ½ inch nominal pipe size (nps) (DN 15). The valve shall withstand seawater inlet pressures up to 200 lb/in^2 (1379 kPa). The valve shall be constructed of nonferrous or corrosion-resisting material. The valve body shall be made of valve bronze alloy C92200 in accordance with ASTM B61 or alloy C95800 in accordance with ASTM B148. The internal metal parts subject to corrosion or erosion shall be made of nickel-copper in accordance with QQ-N-281. The valve disc shall be a conical tapered configuration, proportioning type for smooth actuation and flow regulation under partial load, cold seawater, or high inlet water pressure conditions. The valve shall have either flange connections in accordance with MIL-PRF-20042 or union ends in accordance with MIL-DTL-1183. The valve shall be constructed to prevent the entry of seawater into the refrigerant system in the event of derangement. The water regulating valve shall be located within the air conditioning cabinet so that no refrigerant piping or connections are outside the cabinet enclosure.

3.9.6 Strainer. A strainer in accordance with Drawing 803-841499 shall protect the condenser and water regulating valve. The strainer shall be not less than ¾ inch nps (DN 20). However, it shall be sized so as to not restrict seawater flow during overload conditions. The strainer shall be shipped loose for installation in the seawater supply piping during air conditioner installation.

3.9.7. Expansion valve. Refrigerant flow to the cooling coil (evaporator) shall be controlled by a thermostatic expansion valve. The valve shall be an adjustable superheat, externally equalized type. The valve body shall be of cast or forged brass or bronze and the inlet and outlet connections shall be an integral part of the valve body.

3.9.8 Piping. Any piping necessary for the operation of the equipment shall be provided, up to and including fittings at each unit required for interconnection to supplementary service. Exterior connection fittings shall be capped or plugged to safeguard against damage before installation. Piping connections shall be arranged in such a manner as not to impair the vibration isolation properties of vibration absorption-type mounts. Piping shall be securely supported to minimize strain and vibration and shall be protected against abrasion from adjacent metal parts. Piping connections other than mechanical shall be silver brazed. The water regulating valve shall be installed in the condenser water discharge line and shall be in accordance with MIL-T-16420, alloy C70600 type I, class 200, grade 1. Water pipe fittings and seawater service connections shall be 90-10 copper-nickel, valve bronze in accordance with ASTM B61 or 70-30 copper nickel in accordance with QQ-N-281. No tapered pipe threads shall be used anywhere in the unit. Refrigerant piping shall have readily accessible test fittings for high and low pressure and for charging and draining refrigerant. Refrigerant pipe fittings shall be in accordance with ASTM B16/B16M. Refrigerant piping and evaporator condensate drain tube shall be copper in accordance with ASTM B75/B75M, alloy C12200.

3.9.8.1 Disposable dehydrator. A disposable dehydrator shall be provided in the refrigerant circuit. The dehydrator shall be equipped with an auxiliary screen or other protective means at the dehydrator outlet to prevent passage of the dehydrating agent in the event of rupture of the cartridge screen outlet. The desiccant shall be silica gel, activated alumina, or other material meeting this requirement.

3.9.8.2 Strainer. A fine (80 to 100) mesh strainer shall be located upstream from the expansion valve.

3.9.8.3 Moisture indicator. A combination sight flow moisture indicator shall be installed between the strainer and the expansion valve. The moisture indicator elements shall be replaceable without removing the body from the refrigerant piping.

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3.9.9 Cabinet. The cabinet enclosure shall be constructed of steel or aluminum protected against corrosion. Aluminum shall be ASTM B209, alloy A95052, temper H-32 ¼ hard. Steel shall be low carbon steel, ASTM A1011/A1011M or equivalent. The cabinet shall incorporate framing, chassis, and support for any component of the air conditioner. The cabinet, framing, and chassis shall support and maintain proper alignment and arrangement of every component under the operational conditions as specified in table III and 3.16. The air handling section of the cabinet shall be constructed for front air intake and top air discharge. The air intake shall be provided with a protective grill. The front of the machinery compartment shall be provided with a removable panel for servicing the equipment. The back of the machinery compartment shall be enclosed with an expanded metal screen for service 2 and service 3 air conditioners. Corrosion protection of the cabinet shall be accomplished by cleaning and drying of the metal and application of pretreatment primer in accordance with TT-C-490, zinc chromate primer in accordance with TT-P-645 formula 84, and light gray enamel in accordance with MIL-DTL-15090, class 2. No wood shall be used in the cabinet. Brackets or integral frame members shall be provided for mounting and fastening the air conditioner to a deck foundation. A condensate drip pan shall be provided and shall collect condensate from the cooling coil (evaporator). The pan shall be constructed from corrosion-resisting material. The pan shall have depth and baffles and multiple drain outlets to prevent overflow of the condensate and provide for draining of the condensate under conditions as specified in 3.16. The drip pan outlets shall be interconnected to a common drain tube.

3.9.10 Plenum. An air discharge plenum shall be provided when specified (see 6.2). The plenum shall be detached and constructed to mount on top of the cabinet. The plenum shall contain air guide vanes and louvers as required for directing the air flow. The outlet louvers shall permit adjustable directional air flow in both horizontal and vertical planes. Corrosion protection of the plenum shall be identical to that for the cabinet.

3.9.11 Insulation. Thermal insulation shall be applied as necessary to the cabinet, piping, and components to prevent sweating, dripping, running off, or blowing off of moisture under conditions specified in 4.6.6.2. Acoustical insulation shall be applied to minimize the transmission of noise generated by the air conditioner to surrounding area. For noise criteria, see 3.14 and 3.15.

3.9.12 Air-circulating equipment. Air-circulating equipment shall include a fan, fan motor, drive, motor control, and air filters. The unit shall have the capacity for circulating air through the air conditioning unit at not less than 25 cubic feet per minute (0.7 cubic meter per minute) per 0.8 TOR (1,000 Btus) per hour. The unit shall deliver the amount of air as shown in table III when it is used in conjunction with an external duct system and the additional resistance in the air stream due to this external duct system is that as shown in table III (see 4.6.6.3).

3.9.13 Fans. Fans shall be of the centrifugal type and quiet in operation. They shall be secured to shafts and shall be supported by not less than two self-aligning bearings. Bearings shall be replaceable permanently lubricated ball bearing type in accordance with MIL-B-17931. Bearing tolerances shall be in accordance with ABMA 4. Bearings shall be replaceable without removal of the blower from the front or sides of the unit after removal of the service panel. Fans shall have a V-belt drive. Fan motors shall have variable pitch pulleys and an adjustable base or rail in order to adjust belt tension.

3.9.14 Air filter. Air filters shall be arranged to filter ventilation or recirculated air before it enters the evaporator. Filters shall be of the permanent washable type. Materials used in the construction of the filters shall be corrosion-resisting as specified in 3.4.2 or aluminum. The air filters shall be arranged so they shall not come in contact with condensate from the cooling coils.

3.9.15 Electrical equipment. The electrical equipment shall be in accordance with NEMA MG 1, MIL-DTL-17059, and MIL-M-17060. Electrical equipment shall operate in a 122 °F (50 °C) ambient temperature. No portion of the electrical circuit shall be grounded. The frames or enclosures of all electrical components shall be grounded to the frame of the air conditioning unit to eliminate hazard from shorts or grounds within the equipment. The air conditioner shall be provided with a motor controller for remote mounting (see 3.9.15.2). Wire for interconnection of electrical components shall be in accordance with NEMA HP 3.

3.9.15.1 Motors. All compressor motors shall start the compressor with the maximum refrigerant pressure differential. An unloading device may be used to meet this requirement, if needed. Motors shall be constant speed, continuous duty, with a maximum speed of 3600 r/min.

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3.9.15.1.1 Hermetic motors. Hermetic motors shall be in accordance with NEMA MG 1. The temperature rise of motors in accordance with NEMA MG 1 shall not exceed 158 °F (70 °C) and shall be provided with built-in thermal protectors installed in accordance with NEMA MG 1.

3.9.15.1.2 Other motors. Other motors shall be in accordance with MIL-DTL-17060. Motors shall be service C, constant speed, continuous duty, squirrel cage induction motors for 122 °F (50 °C) ambient temperature, class B or F insulation with winding temperature rise not to exceed 158 °F (70 °C) above ambient and shall use ball bearings in accordance with MIL-B-17931. Bearing tolerances shall be in accordance with ABMA 4. The temperature rise of the outer ring of the bearing shall not exceed 104 °F (40 °C) above ambient. The stator windings shall be insulated through a vacuum pressure impregnation treatment to provide a sealed insulation system. The motors shall be drip-proof or explosion-proof enclosed, as specified in 3.13 (see 6.2). For 440-volt, three-phase motors, the efficiency shall be not less than the value shown in table IV for the size motor provided. Where the motor horsepower (hp) is not listed, the efficiency value for the next lower hp shall be used.

TABLE IV. Motor efficiency.

Motor hp	Efficiency (percent)
15.00	91
10.00	90
7.50	90
5.00	89
3.00	88
2.00	87
1.50	86
1.00	86
0.75	86
0.50	82
0.33	80
0.25	79
less than 0.25	70

3.9.15.2 Motor controllers. Motor controllers shall be constructed to be installed remotely from the air conditioners. The motor controllers shall be in accordance with MIL-DTL-32613.

3.9.15.2.1 Motor controller enclosures. Enclosures for the compressor and fan motor controllers shall be one of the following types defined in MIL-DTL-2036, as specified (see 6.2):

a. Dripproof (45 degrees), watertight, submersible (15 feet)
b. Explosion-proof (class I, group D)
c. Splash-proof

A warning plate shall be mounted on the enclosure with the following engraving in prominent red letters:

WARNING
DANGER HIGH VOLTAGE

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3.9.15.2.2 Motor overload considerations. For hermetic units, overload relays shall be in addition to, and coordinated with, thermal protectors built into the motor. Fans shall be protected with an overload relay or built-in thermal protector. The compressor shall not start or run unless the fan is operating and the compressor shall stop if the fan motor circuit is interrupted. A time delay circuit shall be provided to prevent restarting of the compressor manually or automatically within 5 minutes of having been stopped for any reason. Delay shall not preclude initial starting of the air conditioner.

3.9.15.3 Controls. A selector switch shall permit operation of the fan only, or of the fan and refrigerating equipment. High pressure and low pressure safety switches and a temperature control shall be provided. The selector switch shall be of a three-position type with an off position to secure both the fan and refrigerating equipment. The temperature control shall provide for automatic compressor control. It shall provide for manual adjustment and operate within a temperature range of 65 to 85 °F (18 to 29 °C), with a ±5 °F (±2.8 °C) tolerance. The selector switch shall be readily accessible. The temperature control shall be located within the machinery compartment. The unit shall have separate pressure switches properly set to stop the compressor when the refrigerant discharge pressure rises too high, or suction pressure becomes too low. The pressure controls shall be individual units of the lock open, manual reset type. Pressure and temperature controls shall be in accordance with UL 873. A terminal block or blocks shall be provided with the remote motor controller enclosure (see 3.9.15.2) for the interconnection of all electrical circuits. The internal wiring within the air conditioner cabinet shall be completed to an enclosure containing the necessary terminal blocks, within the cabinet, for interconnection to the motor controller enclosure. Control switches or other electrical devices located external to the motor controller enclosure shall not have protruding or unprotected electrical connection lugs or terminals. Control switches requiring mechanical adjustment shall not be located within the motor controller enclosure, unless provision is made so that the necessary adjustments can be made without removing the enclosure cover. When these control switches are located within the air conditioner cabinet, they shall be mounted, directly or indirectly, by means of mounting brackets or mounting plates, to the cabinet structure to minimize the effect of vibration from the rotating machinery. They shall be located to minimize the possibility of personnel coming in contact with line electrical parts when servicing the controls.

3.10 Identification plates. Identification plates shall be made of brass (type A), aluminum, or corrosion-resisting steel (type H) in accordance with MIL-DTL-15024 and MIL-P-15024/5. Information plates shall be made of laminated plastic (type F), aluminum, or corrosion-resisting steel (type H) in accordance with MIL-DTL-15024 and MIL-P-15024/5. The identification plate shall be located in the front of the machinery compartment. Identification plates shall be secured to equipment with electrolytically compatible fasteners and shall contain the following information:

a. Name of equipment: Air conditioner
b. Manufacturer’s name and address
c. Manufacturer’s model, type, capacity
d. Manufacturer’s serial number
e. Date of manufacture
f. Government or shipbuilder’s contract or purchase order number
g. Specification number and appropriate type, service, class, size (cooling capacity), and power supply
h. Part or identifying number (PIN)
i. National stock number (if available)

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In addition to the identification plate, equipment for the explosion-proof applications shall have a warning plate. The warning plate shall be located on the front of the cabinet. Lettering shall be readily legible, reading as follows:

CAUTION

THIS EQUIPMENT IS FOR EXPLOSION-PROOF APPLICATION. DISCONNECT POWER SUPPLY BEFORE OPENING ELECTRICAL BOX COVERS. DO NOT ENERGIZE UNLESS THE EQUIPMENT IS FULLY ASSEMBLED AND ALL ELECTRICAL ENCLOSURES ARE PROPERLY CLOSED.

3.11 Shock resistance. Unless otherwise specified (see 6.2), each air conditioner and its components shall be shock resistant in accordance with MIL-DTL-901 for grade B, deck mounted, class II, medium weight, type A equipment (see 4.6.1). When specified, grade A shock requirements shall be met.

3.12 Vibration. The air conditioner shall meet type I vibration requirements in accordance with MIL-STD-167-1 (see 4.6.2).

3.13 Drip-proof and explosion proof. The air conditioner shall meet the requirements of MIL-DTL-2036 for dripproof or explosion proof construction as specified (see 4.6.3 and 6.2).

3.14 Airborne noise. The air conditioner and controls furnished under this specification shall meet equipment grade A-3 noise requirements in accordance with MIL-STD-1474 (see 4.6.7).

3.15 Structureborne noise. When specified (see 6.2) the air conditioner and controls shall meet the type III structureborne noise requirements in accordance with MIL-STD-740-2 (see 4.6.7).

3.16 Ship inclination. The air conditioner shall operate on a surface ship while withstanding 15 degrees permanent, or 45 degrees cyclic, inclination (see 4.6.4).

3.17 Capacity. The air conditioner shall deliver not less than its specified capacity when operating at the rating conditions as shown in table III. The net capacity shall be exclusive of all electrical energy required to operate the air conditioner (see 4.6.5).

3.18 Performance. Under rated conditions, the air conditioning unit shall have a minimum performance factor of 9 Btu per hour per watt based on net capacity and total watts input to the unit (see 4.6.6).

3.18.1 Overload. The temperature rise for motors shall not exceed the permissible temperature rise in the applicable motor specification, or 158 °F (70 °C) for motors in accordance with NEMA MG 1 (see 4.6.6.1).

3.18.2 Condensation. The air conditioner shall operate without dripping, running, or blow-off of moisture either inside or outside the cabinet (see 4.6.6.2).

3.18.3 Air delivery. The air flow quantity shall be not less than specified in table III (see 4.6.6.3).

3.19 Pressure. Every refrigerant-containing component shall withstand a pressure of 450 pounds per square inch (3102 kPa) (see 4.6.9).

3.20 Leakage. The refrigerating high side and low side system shall be resistant to leakage when tested in accordance with 4.6.10.

3.21 Water system pressure. The seawater piping system shall not leak seawater at up to 300 pounds per square inch (2068 kPa) (see 4.6.11).

3.22 Insulation motor resistance. The insulation motor resistance of the completed air conditioner shall have a minimum value of 10 megohms at 77 °F (25 °C) (see 4.6.12).

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3.23 Dielectric strength. The electrical equipment and wiring shall withstand the dielectric strength test as specified in 4.6.13.

3.24 Operation. The air conditioner shall operate under the conditions specified herein. Once set, operation shall automatically maintain the environmental conditions for which the equipment is set as specified in table III (see 4.6.14).

3.25 Reliability and maintainability. Each air conditioner shall have an operating life of 100,000 hours with no scheduled maintenance. Corrective maintenance shall be not more than 2 hours per 20,000 hours of operation. Special tools required for maintaining the air conditioner shall be as specified (see 6.2). Special tools are defined as tools not listed in the GSA Global Supply Catalog. Technical manuals shall be as specified (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on air conditioners procured under this specification when a first article sample is required (see 3.1). This inspection shall include the examinations and test listed in table V.

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TABLE V. First article and conformance verifications.

Inspection	Requirement	Test method	First article	Conformance
Visual and dimensional examination	3.4, 3.5, 3.6, 3.8, 3.9, 3.10	4.4	X	X
Shock	3.11	4.6.1	X	--
Vibration	3.12	4.6.2	X	--
Explosion proof (when required)	3.13	4.6.3	X	--
Airborne noise	3.14	4.6.7	X	--
Structureborne noise (when required)	3.15	4.6.7	X	--
Inclination	3.16	4.6.4	X	--
Capacity	3.17	4.6.5	X	--
Performance	3.18 (all)	4.6.6 (all)	X	--
Refrigerant pressure	3.19	4.6.9	X	X
Refrigerant leakage	3.20	4.6.10	X	X
Water pressure	3.21	4.6.11	X	X
Insulation resistance	3.22	4.6.12	X	X
Dielectric	3.23	4.6.13	X	X
Operation and controls	3.24	4.6.14 4.6.15	X	X
Reliability and maintainability	3.25	4.6.8 (all)	X	--
Production reliability	3.25	4.6.16	--	X

4.2.1 Sampling for first article inspection. The first air conditioner of each type, service, and size produced as specified (see 6.2) shall be selected for testing.

4.2.2 First article after test. The first article that has been subjected to the tests and examinations specified in table V and found to be satisfactory may be offered for delivery (see 6.2) subject to the following:

a. Any repairs or modifications found necessary as a result of the first article tests and examinations have been proven satisfactory by retesting.

b. The air conditioner is not degraded in any way.

c. Equipment which has been subjected to the high-impact shock and vibration tests and has successfully passed these tests shall be considered acceptable, subject to the following:

(1) Minor deformation including alignment. (Minor deformations are defined as those which do not cause rejection of the air conditioner under high-impact shock or vibration tests, but which are in excess of the air conditioner dimensional tolerances of the original manufactured design.)

(2) Bearings shall be replaced.

(3) Each part shall be carefully examined for compliance with the original manufactured design and any part which does not meet those original specifications shall be replaced.

(4) Equipment shall be cleaned, leak tested, and dehydrated in accordance with the requirements specified herein.

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(5) Equipment shall be retested in accordance with the conformance tests as specified in table V. Equipment which were subjected to shock and vibrations tests shall, additionally, be tested for capacity in accordance with 4.6.5

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests listed in table V. Production units failing to meet those tests shall be addressed as specified (see 6.2).

4.3.1 Lot. All air conditioners of the same type, service, and size offered for delivery at one time shall be considered a lot.

4.3.2 Sampling for examinations and tests. A random sample of air conditioners shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.4 Visual and dimensional inspection. Each sample unit selected shall be examined for adjustments, fits leaks, material, finish, and general conformance to 3.4, 3.5, 3.6, 3.8, 3.9, and 3.10 and particularly as follows:

a. The external fittings shall be properly secured (see 3.9).

b. Bolts, nuts, and screws shall be tight; equipment and parts shall be properly fastened and secured.

c. No parts shall be fractured, split, torn, dented, or otherwise damaged such as to affect serviceability.

d. There shall be no sharp or ragged edges on the sheeting that may be injurious to personnel.

e. Cold lines shall be properly insulated (see 3.9.11).

f. The limiting and mounting dimensions shall be in accordance with drawings and table II.

g. The temperature control shall be properly set and functioning (see 3.9.15.3).

h. The selector switch shall operate satisfactorily (see 3.9.15.3).

i. There shall not be any dripping, running, or blowing off of moisture (see 4.6.6.2).

j. The low pressure and high pressure switch shall be functioning and properly set in accordance with the manufacturer's specification (see 3.9.15.3).

k. There shall be no refrigerant leakage at any brazed, welded, or mechanical joint as measured using an electronic halide leak detector adjusted to detect a leak of ½ ounce per year.

4.5 First article test procedures. Before testing, the air conditioner shall have successfully passed the visual evaluation specified in 4.4. For first article tests, fresh water may be used in lieu of seawater for cooling. Whereas some tests may be conducted concurrently, the shock test, inclined operation test, and vibration test shall be performed separately. Failure of the air conditioner to pass any of the tests or examinations specified herein shall constitute rejection of the air conditioner.

4.6 Tests.

4.6.1 Shock. One complete air conditioner of each type, service, size, and design shall be subjected to the high-impact shock tests (see 6.3) in accordance with MIL-DTL-901 for grade B or grade A if so specified (see 6.2), except for the following:

a. Resilient mounts shall not be used externally to the air conditioner for mounting the air conditioner to the deck.

b. The air conditioner shall be operating during each shock blow unless it became inoperative during the previous blow. Electric power and seawater shall be applied to the air conditioner during each shock blow even if it has become inoperative. The air conditioner shall be subject to the complete series of shock blows or until it fails the acceptance criteria as specified.

4.6.1.1 Acceptance criteria. The air conditioner shall be considered acceptable for high-impact shock application unless the following conditions occurred during, or were evident at the conclusion of, the test:

a. Dislodgement or displacement of any piece, part, or component in such a manner that would be hazardous to personnel in the vicinity of the air conditioner or cause an electrical short circuit.

b. The shearing or tearing of metallic parts that would create projections or sharp edges.

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c. Where the equipment is constructed to be explosion-proof, any fracture or distortion that reduces the explosion-proof characteristics of the equipment.

d. Any refrigerant or seawater leakage.

4.6.1.2 Guidance. Guidance regarding shock tests and design of shock resistant naval equipment can be found in T9070-AJ-DPC-120/3010 and MIL-DTL-901.

4.6.2 Vibration. One complete air conditioner of each type, service, and size shall be subjected to the type I environmental vibration tests (see 6.3) in accordance with MIL-STD-167-1 (see 3.12). Where the equipment is constructed to be explosion-proof, any deformity resulting from the test that would reduce the explosion-proof characteristics of the equipment shall be cause for rejection.

4.6.3 Drip-proof and explosion-proof. One complete type II air conditioner of each service and size shall be tested to establish conformance to explosion-proof requirements as specified in 3.13. Testing shall be in accordance with MIL-DTL-2036.

4.6.4 Inclination. The unit shall be inclined at an angle of 15 degrees each side of the vertical in each of two vertical plates at right angles to each other and operated at least 1 hour or cycled at 45 degrees for 1 hour, under applied rating conditions as specified in table III and 4.6.5. The unit shall be monitored at 10-minute intervals throughout the test. The unit shall be acceptable if there is no spillage of fluids inside or outside the cabinet, no abnormal noise, and no loss of capacity.

4.6.5 Capacity rating. Capacity rating tests shall be conducted in accordance with the procedures and requirements indicated in ASHRAE 37 and AHRI 210. A standard rating test shall be conducted using standard rating conditions in accordance with AHRI 210. An application rating test shall be conducted using the operating conditions as shown in table III.

4.6.6 Performance. Each air conditioning unit to be tested shall be given the continuous-operating tests as specified in 4.6.6.1 through 4.6.6.3 under the average temperature conditions as indicated with a tolerance of ±1 °F (±0.5 °C) dry bulb, ±0.5 °F (0.25 °C) wet bulb, and ±1 °F (0.5 °C) condenser water temperature. Observations and readings shall be taken at intervals not greater than 10 minutes.

4.6.6.1 Overload. The unit shall be operated with 100 °F (38 °C) dry bulb temperature, 85 °F (29 °C) wet bulb temperature unit-ambient-air and room air entering-air-inlet, and 100 °F (38 °C) water to the condenser. The test shall be continued until steady conditions have been observed for not less than 4 hours. The unit shall operate normally without any interruption caused by tripping of motor-overload devices, without damage to motors due to overheating, and without injury to any other component part from any operational cause. The temperature rise of the compressor motor winding shall be determined at the end of the test. The temperature rise shall not be greater than that specified in 3.9.15.1. The temperature rise of the windings shall be measured and computed by the resistance method. In addition, the air conditioner shall be tested to determine actual capacity under overload conditions as shown in table III.

4.6.6.2 Condensation. The unit shall be operated continuously for 4 hours with 95 °F (35 °C) condenser inlet water, 80 °F (27 °C) dry bulb temperature, and 75 °F (24 °C) wet bulb temperature unit-ambient-air and room air entering-air-inlet. The air conditioning unit shall perform satisfactorily during the test without dripping, running, or blowing off of moisture either inside or outside the cabinet.

4.6.6.3 Air delivery. The unit shall be provided with a means of restricting the outlet air to produce the minimum outlet resistance as specified in 3.18.3. The air flow quantity shall be not less than that specified in table III.

4.6.7 Noise test. The unit shall be tested in accordance with the peak A-weighted sound pressure level measurement procedure and at the measurement locations specified in MIL-STD-1474 (see 3.14). The sound pressure level at each frequency shall be less than or equal to that specified for equipment grade A-3. When specified (see 6.2) type III structureborne noise (see 3.15) shall be tested in accordance with MIL-STD-740-2.

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4.6.8 Reliability and maintainability. The air conditioner shall be disassembled, and all components shall be reassembled and replaced. The personnel and specific types of tools used in the demonstration shall be identified. In addition, the demonstration shall verify that the requirements of 3.3 and 3.25 are satisfied by the design and construction of the air conditioner. The demonstration shall be conducted utilizing the technical manual and the recommended tools without the use of an expert mechanic. The demonstration procedures in MIL-HDBK-470 may be used. Particular attention shall be directed toward the possibility of loss of small parts by the maintenance crew. The reliability validation procedures in MIL-HDBK-217 may be used. The test shall be considered a failure if the air conditioner does not meet the maintainability, reliability, and repair parts requirements of 3.3 and 3.25, or if the air conditioner is unable to be disassembled and reassembled without personnel hazard in accordance with procedures outlined in the technical manual and restored to full operation.

4.6.8.1 Degree of disassembly and reassembly. The degree of disassembly and reassembly shall include, but not be restricted to, the following items, where applicable:

a. Low and high pressure control switches
b. Temperature control switch
c. Sight glass, filters, strainers, screens, and dehydrator
d. Valves, expansion valve, water regulating valve
e. Condenser
f. Fan and fan motor
g. Compressor unit assembly

4.6.8.2 Exceptions. Items of the following nature need not to be performed:

a. Destruction of material such as disassembly of electronic components or removal of condenser tubes
b. Boring, grinding, and other shaping repair procedures
c. Unbrazing connections to remove components

4.6.9 Pressure. Each refrigerant-containing component shall withstand a test pressure of 450 lb/in^2 (3102 kPa) (see 3.19). The test pressure shall be attained by first charging refrigerant into the unit to raise the pressure to 45 lb/in^2 (310 kPa). Dry nitrogen shall be used to raise the pressure to the final test pressure.

4.6.10 Leakage test. The refrigerating high side and low side system shall be tested to determine tightness and resistance to leakage at a minimum gas pressure of 450 pounds per square inch (3102 kPa) (see 3.20). The leak testing equipment shall detect refrigerant leaks of ½ ounce of refrigerant per year.

4.6.11 Water system pressure test. The seawater piping system, including all components, shall be tested at a hydrostatic pressure of 450 lb/in^2 (3102 kPa) (see 3.21).

4.6.12 Insulation resistance test. The motor shall be at ambient temperature. The measuring of insulation shall be made by connecting the three line terminals together, manually closing the line and fan contactors, and measuring the resistance between terminals and ground. The insulation resistance shall be measured with a 500-volt insulation resistance meter. The time of test voltage application shall be not less than 60 seconds (see 3.22).

4.6.13 Dielectric strength test. The electrical equipment and wiring shall withstand, for a period of 1 minute, a dielectric ac, root-mean-square (rms) test voltage or 1900 volts (see 3.23). The test voltage for all other electrical equipment and wiring shall be two times the operating voltage plus 1000 volts.

4.6.14 Operation and controls. Each air conditioner shall be operated for a period of at least 1 hour with controls set to allow continuous compressor operations as shown in table III (see 3.24). It shall be verified that the controls are adjusted and functioning properly. During this test, inlet air temperature to the evaporator shall be not less than 75 °F (24 °C) dry bulb. The electrical power input shall be continuously monitored during this test. At the conclusion of this operating test, the entire refrigerant circuit, including connections under refrigerant pressure shall be tested to determine leakage (see 4.6.10).

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4.6.15 Electrical power. The electrical power input shall be monitored during the operating tests and compared with the input of all other units that have been tested. If any unit requires 7 percent more power than the average of all the acceptable units, it shall be rejected.

4.6.16 Production reliability test. A production reliability test shall be conducted to ensure that the air conditioner meets the requirements of 3.2 (see 6.3). The air conditioner shall be operated for 500 hours with a 24-hour non-operative period every 100 hours of operation. For the conformance inspection, the air conditioner shall be required to operate only for one 100-hour period. The air conditioner shall be operated in all conditions of list, trim, roll, and pitch in accordance with 3.17. The air conditioner shall be in the AUTO mode during the test. This test may be run concurrently with any other applicable test. Production units failing to meet this test shall be addressed as specified (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. These air conditioners are intended to meet Navy special requirements for shipboard use in compartments and areas where central system air conditioning is not provided but seawater for cooling is available. Current stock of air conditioners may be used until depleted; future requisitions for air conditioners should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, service, class, capacity, and size required (see 1.2 and 3.6).
c. When first article inspection is required (see 3.1 and 4.2).
d. Construction, if otherwise specified (see 3.5).
e. When air discharge plenum is required (see 3.2 and 3.9.10).
f. When drip-proof protected or explosion-proof equipment is required (see 3.9.15.1.2 and 3.9.15.2.1).
g. When grade A shock requirements must be met (see 3.11 and 4.6.1).
h. When type III structureborne noise criteria is required (see 3.15).
i. Special tools, if required (see 3.25).
j. Requirements for first article equipment offered for delivery after testing (see 4.2.2).
j. Acceptance Quality Limit (AQL) of 1.0 (see 4.3.2).
k. Requirements for failed production units (see 4.6.16).
1. Packaging requirements (see 5.1).
m. Technical manual requirements (see 6.3).
n. Provisioning requirements (see 6.6).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract (see 6.2). If technical manuals are required, they should comply with MIL-DTL-24784. The technical manuals should be acquired under separate contract line item in the contract.

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6.4 First article inspection. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, samples selected from the first lot size of production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions. Terms and phrases used throughout this specification are defined as follows:

6.5.1 Air discharge plenum. The air discharge plenum is an enclosure (housing) containing guide vanes and a grill or louvers in the outlet opening for directing the airflow.

6.5.2 Air handling/evaporator section. The air handling/evaporator section is an enclosed component consisting of an air filter with drive motor, evaporator (cooling coil), refrigerant flow control, and condensate collector and drain.

6.5.3 Condensing unit. The condensing unit is an assembly of refrigerating components, consisting of a compressor with a drive motor, condenser, receiver, operating and safety controls, and interconnecting piping and wiring. The controls should include a condenser water regulating valve, high and low refrigerant pressure switches, refrigerant relief device, and motor overload protection.

6.5.4 Cooling effect. The net room sensible cooling effect of a unit is defined as the difference between the net total cooling effect and the dehumidifying effect, expressed in Btu per hour. The net total cooling effect is the total useful capacity of the unit for removing heat from the space to be treated, expressed in Btu per hour. The net dehumidifying effect is the difference between the moisture content in pounds per hour of the entering and leaving air, multiplied by 1060, expressed in Btu per hour.

6.5.5 Diamètre nominal (DN). The European and international designation equivalent to NPS in which sizes are measured in millimeters.

6.5.6 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract (see 6.2).

6.7 Subject term (key word) listing.

Air discharge plenum

Compressor

Condensing unit

Evaporator

Liquid receiver

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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